AV AI VELOCITY ARCHITECT INTELLIGENCE · ISSUE 03

WEEK OF 12–18 SEPTEMBER 2026

What The Yes Covered

Is an approval bound to the arguments that run, or to the ones the model wrote? Three frameworks answered three ways this week.

22 repos read

2 mechanisms executed

7 papers read

13 models priced

Vijay Zharotia
Healthcare AI architecture

WHY AI VELOCITY

Agents that act inherit obligations.

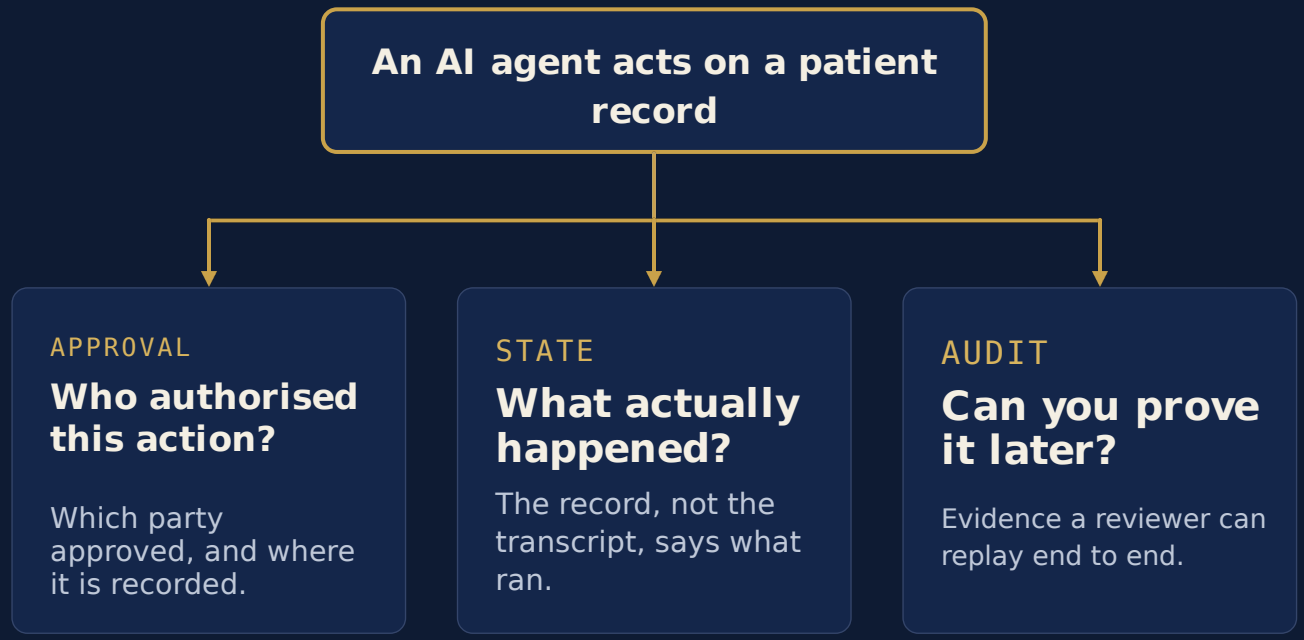
A weekly architecture brief for teams putting AI agents into healthcare workflows. We read the releases, run the mechanism, and say what to change.

FOR ARCHITECTS

The mechanism, the code, the move.

FOR PRODUCT MANAGERS

The decision, the risk, the date.



EVERY CLAIM CARRIES ITS EVIDENCE

EXECUTED

We ran it. The numbers are observed output.

DOCS

Read from documentation or source.

UNVERIFIED

Not confirmed. Vendor numbers carry an asterisk.

THE WEEK IN ONE VIEW

From who answers to what the answer covers

CONTEXT

Last week: who may answer

A clock and a token audience could decide before the clinician. This week the question moved inside the call itself.

→

PROBLEM

The yes covered other arguments

A policy read the model’s JSON while the tool ran validated values. An omitted route became IV, and 1.5 g became 1,500 mg, with no clinician.

→

SOLUTION

Bind the yes to what runs

Approve the validated, executed arguments, and show exactly those to the person who signs.

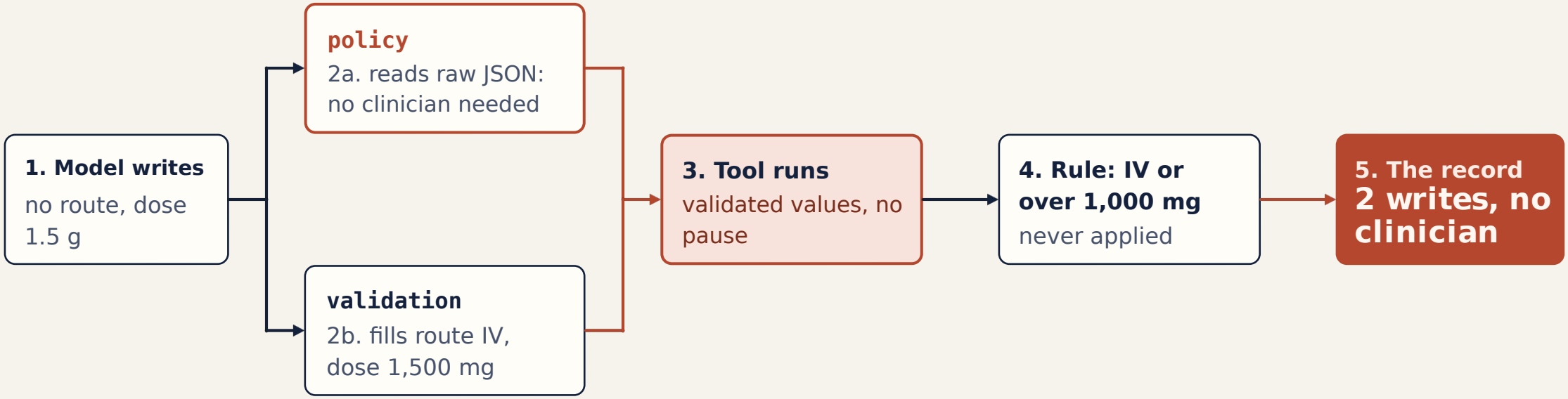
APPROVAL · 04–06
What a yes is bound to

CONTROLS · 07
Settings that did not hold

EVIDENCE · 08
Runs, traces, auditors

COST, RULES · 09–10
Prices and evidence bars

The policy approved arguments it never saw



The policy decided on the model’s JSON; the tool executed the validated arguments. Two different objects.

EXECUTED MECHANISM · SOLUTION · 0.22.3 AND YOUR SCREEN

Show the approver what will actually run

STEP 1

Validate first

On 0.22.3 the policy reads the validated call.

→

STEP 2

Render what will run

Validated arguments, defaults filled in, on the approval screen.

→

STEP 3

Flatten the arguments

Model-typed arguments now pause every call.

→

STEP 4

Then execute

The yes now covers the exact call.

2

writes the policy missed on 0.22.2

0

unapproved writes on 0.22.3

1 of 2

safe calls now paused as well

> The gap moved from the policy to the screen: on 0.22.3 the clinician is still shown the raw JSON, no route and 1.5 g.

Three vendors, three answers to one question

CONTEXT
how it runs

Model proposes a call → **Middleware repairs args** → **Approval is cached** → **Human edits, call runs** → **Audit export**

PROBLEM
where it broke

AGENT FW 1.19
A repaired argument now voids the yes
DOCS

MCP SEP-2640
A remembered yes binds to content digests
DOCS

LANGCHAIN 1.4.2
Transcript says 1000 mg; 500 mg ran
EXECUTED

SOLUTION
what to build

Bind the yes to the call that runs

Repair before policy, key cached approvals on a hash of name, description and schema, and write the audit row from the executed call, never the AIMessage.

Two settings that were quietly ignored

CONTEXT
how it runs



PROBLEM
where it broke

FASTMCP 4.0.5
A StrictInt dose took the string “5”
DOCS

PYDANTIC-AI 2.44
Redacted instructions still reached traces
DOCS

SOLUTION
what to build

Test the control, not the config

Assert that “42” is rejected on every strict field of every write tool, and search FallbackModel and streamed spans since redaction was enabled.

Three tests your own logs cannot pass

CONTEXT
how it runs

Eval passes → **Agent writes orders** → **Per-call policy check** → **Internal audit** → **Write to the record**

PROBLEM
where it broke

2609.13582
43 of 43
order groups differed across five runs

2609.18820
4 patterns
each step passes; the trace breaks policy

2609.18272
5.9%
of visible faults found by internal audit

SOLUTION
what to build

Check across runs, traces and stacks

Repeat each eval five times and fail on any change in orders, keep running totals per patient across the trace, and store audit evidence where the agent cannot write.

COST · LIST PRICES READ 20 SEPTEMBER

A correction, and a cheaper floor

MODEL	COST PER 10,000 DOCS	INDEX
GLM-5.3-Flash	**$8.50**	**42**
Qwen3.8-27B	**$13***	**22***
DeepSeek V4.1 Flash	**$18**	below 42
GPT-5.6 Sol	**$260 promo**	**47**
Claude Opus 5	**$325**	**51**

Per million tokens in/out: GLM $0.15/$0.50, DeepSeek $0.30/$1.20, Qwen $0.10/$1.80 on OpenRouter, Sol $4/$20 promo, Opus 5 $5/$25. 4,000 in and 500 out per document. Index = Artificial Analysis Intelligence v4.3. *Qwen as read 20 Sep; AA now lists 27B (Low) at 26 and $0.50/$3.00.

CONTEXT

Issue 02 printed Sol at $350, saying the promotion had ended at $5/$30.

CORRECTION

It had not. OpenAI lists $4/$20 through at least 21 November.

DECISION

Reverse a move off Sol only with the November expiry in the budget. Keep score, read date and price expiry as one record.

CONSTRAINTS · CONTRACTS AND PAYMENT RULES

Two rules that set the evidence bar

CMS · ACCESS MODEL · 15 SEP

Medicare will pay on outcomes for AI-enabled chronic care

New tracks start spring 2027: heart failure, COPD, substance use disorder and tobacco cessation. Payment rests on specified outcomes; no data standard is named.

Move. Design the outcome measurement as an audited pipeline now.

ANTHROPIC · COMPLIANCE API

A vendor transcript is not your record of a FHIR bundle

Tool inputs in compliance transcripts are capped at 10,000 bytes by default*. Managed Agents are not eligible for zero data retention or BAA coverage.

Move. Keep your own full-fidelity tool log for PHI workflows.

*As read 18 September; the 10,000-byte default was not found in Anthropic’s public docs on 17 October.

Everyone talked evidence; nobody published a format

EPIC

Agent Factory build-a-thon

25 health systems built refill and med-reconciliation agents. No published write-back audit contract.

LANGCHAIN

Queues as clinical oversight

Included Health runs federated agents on Deep Agents, with annotation queues as the review layer.

OPENAI

Reporting misalignment

A framework covering unauthorised action and concealment, with six disclosed cases.

WILLISON

Logs that could not explain

An operator’s own logs could not reconstruct what its agents did to RubyGems in May.

AIUC

Certification insurers price

AIUC-1 pairs third-party adversarial testing with evidence specifications.

QUIET THIS WEEK

LangGraph and Semantic Kernel releases, HL7, FDA on AI, ONC, NIST, the MCP blog, the Anthropic engineering blog.

Seven moves, with what forces each

Move	Why	By
Show validated arguments, defaults filled, on the approval screen	Approver sees raw JSON	This sprint
Write the audit row from the executed call, not the AIMessage	LangChain 1.4.2 split them	Before upgrading
Put argument repair before policy middleware	Repair voids the approval	Next upgrade
Key cached approvals on a hash of name, description, schema	MCP SEP-2640	Next release
Test that “42” fails every strict field on write tools	FastMCP coerced strict fields	This sprint
Search FallbackModel and streamed spans since redaction	PHI may have reached traces	Now
Budget Sol with its promo end written in	$4/$20 to at least 21 Nov	21 Nov

AV AI VELOCITY

THIS WEEK'S POSITION

What the approver reads is the line an auditor will ask about.

Long-form edition
Every mechanism, source and date

Lab package
Runs clean, no API key needed

New issue every Friday
Follow AI Velocity on LinkedIn

github.com/zkumar/ai-velocity-weekly